

Erosive potentials of brewed teas.

PURPOSE: To measure the pH, titratable acidity, fluoride concentration and erosive potential of brewed teas. METHODS: Bag teas were purchased to represent black, green, citrus, fruity, and floral tea flavors from Tulsi, Bigelow, HyVee, Tazo, and Yogi brands and brewed (1 bag/240 ml) in boiling water for 3 minutes. The pH, titratable acidity, and fluoride concentrations were measured. Following these measurements, a representative tea from each flavor was selected for investigation of erosion potential. Six extracted human molars were randomly assigned to each tea. Teeth were painted with fingernail polish to expose a 1 x 4 mm window and then soaked in tea for a total of 25 hours with teas refreshed every 5 hours. Teeth were then sectioned using a microtome and photographed using a polarized light microscope. Lesion depths (i.e., eroded surfaces) were measured using Image Pro Plus software. Differences in physiochemical properties and lesion depths between beverages were investigated using one-way ANOVA with post-hoc Tukey's HSD test. Relationships among lesion depths and physiochemical properties were evaluated using the Pearson correlation test. RESULTS: pH, titratable acidity and fluoride concentrations differed between tea flavors ($P < 0.05$) and between brands ($P < 0.05$). Lesion depths produced by the citrus tea (83.1 ± 10.3 microm) were greater than those produced by the fruity tea (56.5 ± 6.1 microm); both teas produced greater depths than black (30.1 ± 7.4 microm), floral (25.0 ± 3.2 microm) or green (22.3 ± 6.3 microm) teas ($P < 0.05$). pH ($r = -0.96$; $P = 0.009$) was inversely and titratable acidity ($r = 0.97$; $P = 0.006$) was positively associated with lesion depths.